

Methylmercury Exposure and Incident Diabetes in U.S. Men and Women in Two Prospective Cohorts

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OBJECTIVE Emerging in vitro and animal evidence suggests that methylmercury could increase type 2 diabetes, but little evidence exists in humans. We aimed to prospectively determine associations of mercury exposure, as assessed by biomarker measurement, with incident diabetes.

RESEARCH DESIGN AND METHODS We used neutron activation analysis to measure toenail mercury, an objective biomarker of methylmercury exposure, in 9,267 adults free of diabetes at baseline in two separate U.S. prospective cohorts. Incident diabetes was identified from biennial questionnaires and confirmed by validated supplementary questionnaire using symptoms, diagnostic tests, and medical therapy. Associations of mercury exposure with incident diabetes were assessed using Cox proportional hazards.

RESULTS During mean $\pm$ SD follow-up of 19.7 ± 7.0 years, 1,010 new cases of diabetes were diagnosed. The 95th percentile of toenail mercury was $1.32 \mu\text{g/g}$ in men and $0.76 \mu\text{g/g}$ in women, corresponding to exposures ~ 3.5 -fold and 2-fold higher than the U.S. Environmental Protection Agency reference dose. In multivariable analyses, toenail mercury concentrations were not associated with higher incidence of diabetes in women, men, or both cohorts combined. Comparing the highest to lowest quintile of exposure, the hazard ratio (95% CI) for incident diabetes was 0.86 (0.66–1.11) in women, 0.69 (0.42–1.15) in men, and 0.77 (0.61–0.98) in the combined cohorts. Findings were similar when more extreme categories (deciles) of mercury were compared, and in analyses stratified by fish or omega-3 consumption, BMI, and age.

CONCLUSIONS These findings from two separate large prospective cohorts do not support adverse effects of methylmercury on development of diabetes in men or women at usual levels of exposure seen in these populations.